



RV College of Engineering®
Mysore Road, RV Vidyanikethan Post,
Bengaluru - 560059, Karnataka, India

Question Paper ID: 557779

Course Code: HS271TA

USN

1	R	V	2	2	C	5	0	6	7
---	---	---	---	---	---	---	---	---	---

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

Common to all Programs

Indian Knowledge System

Time : 1.5 Hours

Maximum Marks : 60

Instructions to the students

1. Answer all questions from Part A.
2. Encircle the correct answer in OMR sheet only.

Part A

Question No	Question	M	CO	BT
1.1	The motto <i>"Satyam vada, dharmam chara"</i> comes from: a) Taittiriya Upanishad b) Bhagavad Gita c) Arthashastra d) Rigveda	01	1	1
1.2	The term <i>"Indian Knowledge System (IKS)"</i> refers to: a) Western interpretations of Indian culture b) Indigenous systems of knowledge evolved in India over millennia c) Only Vedic scriptures d) Modern Indian scientific research	01	1	1
1.3	In Indian Knowledge Systems, <i>Pratyakṣa</i> refers to: a) Logical inference b) Scriptural testimony c) Direct perception d) Ethical conduct	01	1	1
1.4	Which combination correctly represents kinds of traditional knowledge? a) Ritualistic and industrial b) Ecological, technical, and social c) Commercial and digital d) Political and military	01	2	1
1.5	Western education in India primarily emphasized: a) Oral transmission b) Integrated learning c) Discipline-centric knowledge d) Guru-Śiṣya system	01	1	1
1.6	<i>"Takshashila"</i> and <i>"Nalanda"</i> were: a) Vedic rituals b) Ancient centers of higher learning c) Holy temples d) Trade centers	01	1	1
1.7	Which text belongs to the Vedic Corpus? a) Purāṇas b) Upaniṣads c) Smṛtis d) Itihāsas	01	1	1
1.8	The concept <i>"Sa Vidya Ya Vimuktaye"</i> means: a) Knowledge is wealth b) Knowledge leads to liberation c) Education gives power d) Learning is endless	01	1	1
1.9	Indian philosophical systems differ mainly because they: a) Reject debate b) Share identical metaphysics c) Encourage multiple viewpoints d) Focus only on rituals	01	2	1

- 1.10 The language predominantly used for transmitting knowledge in ancient India was: 01 1 2
a) Pali b) Tamil c) Sanskrit d) Prakrit
- The traditional Indian system of education was centered around:
- 1.11 a) Gurukula system b) British schooling system c) Online learning d) Modern universities 01 1 2
- The primary mode of learning in Gurukulas was:
- 1.12 a) Debate and memorization (*Shravana*, *Manana*, *Nididhyasana*) b) Written examinations c) Multiple-choice tests d) Internet-based study 01 1 2
- 1.13 Piṅgala's binary ideas emerged primarily from his study of: 01 2 1
a) Astronomy b) Algebra c) Geometry d) Chandas
- 1.14 How many sūtras are there in Pāṇini's Aṣṭādhyāyī? 01 2 1
a) 3000 b) 3500 c) 3983 d) 4200
- 1.15 In Sanskrit grammar, suffixes used for noun declensions are known as: 01 2 2
a) Tiṅ b) Dhātu c) SuP d) Vārtika
- 1.16 In Sanskrit grammar, the basic element from which words are generated is called: 01 2 2
a) Pada b) Dhātu / Prātipadika c) Vākya d) Samāsa
- 1.17 In Sanskrit phonetics, *Sthāna* refers to: 01 2 2
a) Effort of pronunciation b) Tone of voice c) Place of articulation d) Meaning of sound
- 1.18 The Bhūtasankhyā system represents numbers using: 01 2 2
a) Only digits b) Alphabets with place value c) Objects, beings, or concepts d) Binary notation
- 1.19 In the Kaṭapayādi system, numbers are encoded mainly to: 01 2 2
a) Encrypt secret messages b) Perform calculations faster c) Aid memorization through meaningful words d) Replace the decimal system
- 1.20 Which Pramāṇa involves understanding a new object through similarity with a known one? 01 2 1
a) Pratyakṣa b) Anumāna c) Upamāna d) Śabda
- 1.21 Which of the following is a *means of valid knowledge (Pramāṇa)*? 01 2 1
a) Prameya b) Saṃśaya c) Pratyakṣa d) Saṃskāra
- 1.22 In the Nyāya system, Anumāna (inference) is structured using how many logical steps? 01 2 1
a) Three b) Four c) Five d) Seven
- 1.23 The arrangement of Piṅgala's *Ganas* corresponds in modern computer science to which concept? 01 2 2
a) Finite automaton b) Binary tree c) De Bruijn sequence d) Huffman coding
- 1.24 In Piṅgala's system, *Laghu* and *Guru* syllables correspond respectively to: 01 2 1
a) 0 and 1 b) 1 and 0 c) 1 and 2 d) Short and long meters only
- 1.25 According to Vedāṅga Śāstras, mathematics is compared to which of the following? 01 3 1
a) Roots of a tree b) Crown of a king c) Crest on the head of a peacock d) Wings of a bird
- 1.26 Which system uses consonants to represent numerical values? 01 3 1
a) Bhūtasankhyā system b) Binary system c) Kaṭapayādi system d) Decimal system

- 1.27 Which Indian mathematician formulated rules for zero and negative numbers? 01 3 1
a) Āryabhaṭa b) Brahmagupta c) Bhāskara I d) Madhava
- The Bhūtasāṅkhyā system represents numbers using:
1.28 a) Only digits b) Algebraic symbols c) Names of objects and concepts 01 3 2
d) Roman numerals
- Which text is considered the primary source of Vedic astronomy?
1.29 a) Sūrya Siddhānta b) Āryabhaṭīya c) Vedāṅga Jyotiṣa 01 1 1
d) Brahmasphuṭasiddhānta
- The Naksatra system in Indian astronomy consists of :
1.30 a) 12 constellations b) 24 constellations 01 3 1
c) 27 lunar mansions d) 30 lunar mansions
- The concept that Earth rotates on its axis causing day and night was
1.31 proposed by: 01 3 1
a) Varāhamihira b) Brahmagupta c) Āryabhaṭa d) Madhava
- The Panchāṅga consists of how many elements?
1.32 a) Three b) Four c) Five d) Seven 01 3 1
- Which of the following is NOT a limb of the Panchāṅga?
1.33 a) Tithi b) Vara c) Yoga d) Rāśi 01 3 2
- In the lost-wax process, the wax model is primarily made from:
1.34 a) Resin b) Tree gum c) Beeswax d) Lac 01 3 1
- The yantra used to provide uniform heating using sand is:
1.35 a) Dola Yantra b) Patana Yantra c) Valuka Yantra d) Dhupa Yantra 01 3 1
- Kapha Doṣa is dominantly composed of which Mahābhūtas?
1.36 a) Air and Ether b) Fire and Water c) Water and Earth 01 3 2
d) Fire and Air
- Improving sleep and nutrition primarily enhances which component first?
1.37 a) Consciousness b) Intellect c) Body d) Sattva 01 3 2
- Kapha Doṣa, when imbalanced, is most closely associated with which Guṇa?
1.38 a) Rajas b) Tamas c) Sattva d) None 01 3 2
- Meditation and ethical living primarily help in increasing:
1.39 a) Rajas b) Tamas c) Kapha d) Sattva 01 3 2
- Which sequence correctly represents information flow in BMI-C?
1.40 a) Body → Mind → Intellect b) Mind → Body → Consciousness 01 3 2
c) Senses → Mind → Intellect → Action
d) Consciousness → Mind → Body
- Architecture (Vāstu Vidyā) is classified under which category of Kalas?
1.41 a) Performing arts b) Nature sciences 01 1 2
c) Engineering and craftsmanship d) Games and sports
- The classical list of 64 Kalas represents:
1.42 a) Religious rituals b) Military sciences c) Holistic life skills 01 1 2
d) Only fine arts
- Knowledge of regional languages belongs to which Kala group?
1.43 a) Performing arts b) Literary and intellectual arts 01 1 2
c) Domestic skills d) Martial arts

1.44	Vāstu Śāstra primarily aims at aligning buildings with: a) Political systems b) Modern technology c) Cosmic and natural forces d) Economic efficiency	01	1	2
1.45	Soil testing and land analysis fall under which limb of Vāstu? a) Prāsāda b) Vāstu c) Yāna d) Śayana	01	1	2
1.46	The Vāstu Puruṣa Maṇḍala is mainly used for: a) Decorative carving b) Structural strength c) Site energy zoning d) Cost estimation	01	1	2
1.47	Which of the following is NOT an limb of Vāstu Śāstra? a) Sthāpatya b) Prāsāda c) Yantra d) Ābharaṇa	01	1	2
1.48	Sustainable consumption-production patterns in IKS relate closely to: a) SDG 6 b) SDG 12 c) SDG 1 d) SDG 14	01	1	1
1.49	Arthaśāstra was written by: a) Vishnugupta / Kautilya b) Patanjali c) Panini d) Kalidas	01	1	1
1.50	Which SDG directly aligns with traditional Indian emphasis on environmental balance? a) SDG 8 b) SDG 13 c) SDG 16 d) SDG 10	01	1	1
1.51	What is the nakshatra in which Sri Ramachandra was born? a) Revathi b) Rohini c) Uttara d) Punarvasu	01	1	1
1.52	Rāma's ability to listen to citizens' grievances reflects: a) Judicial activism b) Participatory governance c) Patrimonial rule d) Neutral administration	01	2	1
1.53	Inspection, espionage, and audit systems in Arthaśāstra relate to: a) Nationalism b) Preventive vigilance c) Ritual governance d) Military conquest	01	2	1
1.54	Which of the following is NOT one of the Saptāṅga of the state? a) Janapada b) Danda c) Kośa d) Yajña	01	2	1
1.55	Which institution in India focuses on protection of traditional healing knowledge? a) NITI Aayog b) Ministry of AYUSH c) RBI d) Election Commission	01	3	1
1.56	Which of the following is incorrect. i) TK is distinctly associated with an indigenous or local community, which preserves and transmits it from generation to generation. ii) TK is old and static. iii) TK is generated, preserved and transmitted in a traditional and intergenerational context. iv) TK is not limited to any specific field.	01	3	1
1.57	Traditional ecological knowledge in India most strongly aligns with which practice? a) Excessive chemical farming b) Sacred groves and community forests c) Mining-based development d) Industrial monoculture	01	3	1
1.58	The term "Daṇḍa" in Arthaśāstra refers to: a) Agricultural surplus b) Law, order, and punitive authority c) Trade networks d) Cultural norms	01	4	1
1.59	Janapada primarily refers to: a) Army b) Territory and population c) Royal palace d) Trade officials	01	4	1
1.60	According to Arthaśāstra, the foremost duty of the king is: a) Expansion of trade b) Promotion of arts c) Protection of subjects (Rakṣaṇa) d) Construction of monuments	01	1	1



RV College of Engineering®
Mysore Road, RV Vidyanikethan Post,
Bengaluru - 560059, Karnataka, India

Question Paper ID: 257011

Course Code: HS271TA

USN

1	R	V	2	2	C	S	0	6	7
---	---	---	---	---	---	---	---	---	---

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

Common to all Programs

Indian Knowledge System

Time : 1.5 Hours

Instructions to the students

Maximum Marks : 40

1. Answer FIVE full questions in the answer booklet given.
2. Answer any one full question from 1 and 2, 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Unit-1

Question No	Question	M	CO	BT
1a	Explain any four differences between vedas and vedangas	04	3	3
1b	Discuss the historicity of Indian Knowledge Systems.	04	1	1

OR

2a	What are Vedas ? List the four vedas	04	2	2
2b	List four major text types or four portions of Vedas. Explain each briefly	04	2	3

Unit-2

Question No	Question	M	CO	BT
3a	Discuss why Pāṇini is often regarded as the "father of linguistics" from both traditional and modern perspectives.	04	2	2
3b	Explain the development of the <i>decimal place-value system</i> in ancient India and its global significance.	04	2	2

OR

4a	Explain <i>Anumāna Pramāṇa</i> with reference to the Pañca-Avayava (five-step inference) structure.	04	2	3
4b	Describe how Piṅgala's arrangement of Ganas relates to the modern concept of the <i>de Bruijn sequence</i> .	04	2	5

Unit-3

Question No	Question	M	CO	BT
5a	Briefly discuss the contributions of any four ancient Indian mathematicians.	04	3	2
5b	Describe the art of perfume making in ancient India and its cultural significance.	04	3	2
OR				
6a	Explain the concept of Panchāṅga and its five elements.	04	3	2
6b	Describe the metallurgical knowledge reflected in the Delhi Iron Pillar.	04	3	1

Unit-4

Question No	Question	M	CO	BT
7a	How does Pitta Doṣa influence physiological activities?	04	3	2
7b	Differentiate between Nāgara and Drāviḍa temple styles.	04	3	3
OR				
8a	Apply the concept of 64 Kalas to modern holistic education	04	3	3
8b	How can Vāstu Śāstra principles be applied in contemporary building design?	04	3	3

Unit-5

Question No	Question	M	CO	BT
9a	Discuss the role of advisors in Rāma's administration.	04	1	3
9b	Name one SDG aligned with Ayurvedic practices.	04	2	1
OR				
10a	Discuss the role of TKDL in safeguarding traditional knowledge.	04	3	3
10b	What are the sources of income for a state as per Arthasastra?	04	4	2